

Human ovarian cancer stem-like cells can be efficiently killed by $\gamma\delta$ T lymphocytes

Dongmei Lai · Fangyuan Wang · Yifei Chen ·
Chunhui Wang · Sha Liu · Bufeng Lu · Xirui Ge ·
Lihe Guo

Received: 28 July 2011 / Accepted: 15 November 2011 / Published online: 27 November 2011
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Abstract Ovarian cancer comprises a small population of cancer stem cells (CSCs) that are responsible for tumor maintenance and resistant to cancer therapies, it would be desirable to develop a therapy that could selectively target ovarian CSCs. Recently, cellular immune-based therapies have improved the prognosis of cancer patients clinically. In this study, we isolated a subset of ovarian cancer sphere cells that possess CSC properties and explored the cell cytotoxicity of $\gamma\delta$ T cells to ovarian cancer sphere cells using a transwell cocultured cell system. The proliferation rate of the cancer sphere cells decreased to 40% after cocultured with $\gamma\delta$ T cells. The $\gamma\delta$ T cells increased the sensitivity of SK-OV-3 sphere cells to chemotherapeutic drugs. After the treatment of $\gamma\delta$ T cells, the expression of stem cell marker genes decreased in sphere cells, while the expression of HLA-DR antigen on tumor cells was increased in a time-dependent manner. Further, $\gamma\delta$ T cells induced G2/M phase cell cycle arrest and subsequent apoptosis in SK-OV-3 sphere cells. Xenograft mouse models demonstrated that $\gamma\delta$ T cells dramatically reduced the tumor burden. Notably, the level of IL-17 production significantly increased after cocultured with $\gamma\delta$ T cells. We conclude that $\gamma\delta$ T cells may efficiently kill ovarian CSCs through IL-17 production and represent a promising immunotherapy for ovarian cancer.

Keywords Ovarian cancer stem cells · $\gamma\delta$ T cells · Immunotherapy · Transwell coculture · Cytokine production

Introduction

Epithelial ovarian cancer remains the most lethal gynecologic malignancy despite recent advances in adjuvant chemotherapy. Although the majority of patients with Stage III/IV ovarian cancer initially respond to platinum-based therapies, recurrence rates are high due to resistance to further treatment with chemotherapy [1]. The mechanisms underlying chemoresistance in cancer are not clear. One hypothesis is that cancers are driven by a subset of highly tumorigenic cells with stem cell properties, cancer-initiating/cancer stem cells (CSCs). Such CSCs have been thought to contribute to chemotherapy resistance [2]. Thus, treatment of the drug-resistant CSC fraction may be critical for ultimate cure rates.

CSCs have been identified in established ovarian cancer cell lines as well as samples from ovarian cancer patients [3, 4]. We have previously obtained self-renewing and anchorage-independent spheroids by culturing patient-derived ovarian cancer cells or SK-OV-3 cell line in stem cell-selective conditions. The spheroid cells expressed stem cell markers including Oct-4, nanog, nestin, Sox-2, ABCG2, CD133 and CD117 surface markers and were more drug resistant. In addition, transplantation of as few as 100–500 such spheroid cells resulted in the formation of tumors that recapitulated the heterogeneous population of the original tumor. In contrast, inoculation of more than 10^5 differentiated cancer cells failed to form tumors [5, 6].

Recent clinical studies have shown that chemotherapy combined with immunotherapy has survival benefits in

D. Lai (✉) · F. Wang · Y. Chen
The International Peace Maternity and Child Health Hospital,
School of Medicine, Shanghai Jiaotong University,
Shanghai 200030, China
e-mail: laidongmei@hotmail.com

C. Wang · S. Liu · B. Lu · X. Ge · L. Guo
Hehong (Shanghai) Biotechnology Ltd., Shanghai, China

comparison with chemotherapy alone [7]. Moreover, chemotherapeutic agents can sensitize tumors to immune cell-mediated killing. For instance, sensitivity of tumor cells to subsequent cytotoxicity is increased by T cells via up-regulation of death receptors DR5 and Fas, to which TRAIL and CD95L/FasL ligands bind, respectively [8]. Most current immunotherapeutic approaches aim at inducing tumor antigen-specific response stimulating the adaptive immune system, which is dependent on MHC-restricted T cells. However, loss of MHC molecules is often observed in cancer cells, rendering tumor cells resistant to T-cell-mediated cytotoxicity [9, 10]. $\gamma\delta$ T cells exhibit potent MHC-unrestricted lytic activity against different tumor cells in vitro, suggesting their potential utility as anticancer therapy. Moreover, $\gamma\delta$ T cells have been consistently identified and isolated from tumor-infiltrating lymphocytes in various types of cancer [11].

Both experimental and clinical studies suggest that $\gamma\delta$ T cells play a significant role in the control of tumor cell development. Activated $\gamma\delta$ T cells exert broad cytotoxic activity toward many different tumor cell types [12, 13]. Thus, $\gamma\delta$ T cells may be used as effectors in tumor immunotherapy. Human $\gamma\delta$ T cells can be selectively activated by nonpeptide antigens or agents such as aminobisphosphonates (N-BP) or monoethyl phosphate (MEP) [14, 15]. MEP mimics small natural ligand and stimulates the proliferation of $\gamma\delta$ T cells in vitro. Previously, we found MEP combined with IL-2 could selectively enhance the proliferation of human $\gamma\delta$ T cells [15].

In this study, we demonstrate the ability of $\gamma\delta$ T cells to efficiently kill ovarian cancer stem-like cells in vitro and in vivo. We also show that the cytokine productions by $\gamma\delta$ T cell are critical for the antitumor effects in ovarian cancer stem-like cells.

Materials and methods

Ovarian CSC culture

The SK-OV-3 ovarian cancer cell line was obtained from Shanghai Cell Bank of Chinese Academy of Sciences and was maintained in McCoy's medium (Sigma-Aldrich, St Louis, USA) supplemented with 10% fetal bovine serum (FBS). Cells were incubated at 37°C in a humidified atmosphere containing 5% CO₂. After growing to 80% confluence, these cells were dissociated by 0.02% trypsin–EDTA for 1–2 min at 37°C and maintained under stem cell conditions by serum-free MEM/F12 supplemented with 5 mg/ml insulin (Sigma-Aldrich), 10 ng/ml human recombinant epidermal growth factor (EGF, Invitrogen, Carlsbad, USA), 10 ng/ml basic fibroblast growth factor (bFGF, Invitrogen) and 0.3% bovine serum albumin (BSA, Sigma-Aldrich). In

this condition, cancer cells grow as nonadherent spheres. Culture media were changed every 2 days by centrifuging at 800 rpm for 5 min to remove the dead cell debris. Regular cell culture plates were coated with poly (2-hydroxyethyl methacrylate) (Sigma) before sphere cells culturing.

$\gamma\delta$ T-cell culture

To expand human $\gamma\delta$ T cells, the peripheral blood mononuclear cells (PBMC) were cultured in RPMI 1640 supplemented with 10% FCS, streptomycin (100 U/ml) and penicillin (100 U/ml), in the presence of 1 mmol/l MEP (Tokyo Kasei) plus recombinant human IL-2 (40 IU/ml; Huaxin, Shanghai). After 14 days of culture, cells were harvested, and purified populations of $\gamma\delta$ T cells were obtained using a human anti-TCR $\gamma\delta$ MicroBead Kit (Miltenyi Biotec). Cell viability was determined using trypan blue exclusion.

Transwell coculture

SK-OV-3 sphere cells were plated in 24-well dishes. $\gamma\delta$ T cells were plated in the transwell chambers (Corning, NY, USA), and these inserts were placed on top of the wells containing the stem cell culture. The membrane in the transwell had a 0.4- μ m pore size that prevents both cell–cell contact and cell migration but allows the diffusion of soluble factors. After 0–6 day's treatment, the SK-OV-3 sphere cells' survival was determined by MTT assay (see below). This coculture system was used for all evaluation described in this study.

Proliferation and drug resistance assessment

As described above, $\gamma\delta$ T cells were grown together with SK-OV-3 sphere cells to obtain S:T (SK-OV-3 cell/T cells) ratios of 10:1 and 20:1. Six days later, sphere cells were collected. For assessment of drug resistance, SK-OV-3 sphere cells cocultured with $\gamma\delta$ T cells were plated in culture medium containing cisplatin (40 μ Mol) and paclitaxel (10 μ Mol) (Sigma) for 48 and 96 h [5, 6]. Cultures were set up in triplicate. The proliferation conditions were determined by MTT assay and the OD reading at 490 nm. The percentage inhibition rate was calculated as follows: $1 - (\text{sample OD}_{490} - \text{blank control OD}_{490}) / (\text{control OD}_{490} - \text{blank control OD}_{490})$.

RNA extraction and real-time qPCR analysis

Total RNA was extracted from SK-OV-3 sphere cells treated with or without $\gamma\delta$ T cells using the RNeasy Mini Kit (QIAGEN, Valencia, USA). Five hundred nanograms of total RNA from each sample was utilized for reverse

Table 1 PCR primer sequences

Gene product	Forward (F) and reverse (R) primers (5' → 3')	Size (bp)
Oct-4	F: GGCCCGAAAGAGAAAGCGAACC R: ACCCAGCAGCCTCAAAATCCTCTC	224
Sox-2	F: GCGCGGGCGTGAACCAG R: CGGCGCCGGGAGATACA	396
Nanog	F: TTCCTTCTCCATGGATCTG R: TCTGCTGGAGGCTGAGGTAT	213
ABCG2	F: TGAGCCTTTGGTTAAGACCG R: TGGTGTTCCTTGTGACACTG	107
TopoIIa	F: GCCGAGTGGAACTTTTGTC R: GTTCATGTGCGCGTAACTGT	104
TopoIIb	F: GAGTGGCTTGTGGGAATGTT R: TGTGCTTCTTTCCAGGCTTT	179
HLA-DR	F: CAGTTCCTCGGAGTGGAGAG R: CTCAGCATCTTGCTCTGTGC	115
B71	F: AAGTATATGGGCCGCACAAG R: CATTCTGTGGGCTTTTGT	120
B72	F: GCACATCTCATGGCAGCTAA R: CACAGGAGCAAGGTTTGTGA	106
18sRNA	F: CGGCGACGACCCATTCGAAC R: GAATCGAACCCTGATCCCCGTC	99

transcription using the iScript cDNA synthesis kit (Bio-Rad, Hercules, USA). Real-time PCR was carried out on cDNA using IQ SYBR Green (Bio-Rad) with Mastercycler ep realplex real-time PCR system (Eppendorf, Hamburg, Germany). All reactions were performed in a 25 µl volume. The primers for the marker genes are provided in Table 1. PCR was performed by an initial denaturation at 95°C for 5 min, followed by 40 cycles for 30 s at 95°C, 30 s at 60°C and 30 s at 72°C. PCR without the template DNA was used as a negative control. Specificity was verified by melting curve analysis and agarose gel electrophoresis. The threshold cycle (Ct) values of each sample were used in the post-PCR data analysis. 18S RNA was used as an internal control for mRNA-level normalization.

Flow cytometry

The expression of a panel of CD133, CD117 and CD44 markers was evaluated on cells obtained from SK-OV-3 sphere cells or from SK-OV-3 cells. 1×10^6 cells were suspended in 2% BSA/PBS and labeled with anti-CD133 (Cell signaling, USA), anti-CD117 (Boshide, P.R. China), anti-CD44 (Boshide, P.R.China) and (Rodamine-labeled, FITC-labeled and cy5-labeled) secondary antibodies. Isolation of CD133⁺, CD117[−] or CD133⁺CD117[−] cells was performed using a FC500 flow cytometer (Beckman Coulter) and analyzed by Beckman Coulter CXP software.

SK-OV-3 sphere cells (1×10^5) treated with or without $\gamma\delta$ T cells were suspended in hypotonic solution (0.1% Triton X-100, 1 mM Tris-HCl, pH 8.0, 3.4 mM sodium citrate, 0.1 mM EDTA) and stained with 50 mg/ml of PI. The DNA content was analyzed by flow cytometry based on the above methods.

Apoptosis was quantified by annexin V-FITC and PI-double staining by using a staining kit (Mbschem, China). The phosphatidylserine was determined by flow cytometry. Briefly, cells were washed twice with PBS and 400 µl binding buffer and 5 µl of annexin V-FITC was added. Following gentle vortex, the mixture was incubated for 15 min at cold room (2–8°C) in the dark and 10 µl PI was added. Following gentle vortex, the sample was analyzed using FC500 flow cytometer within 1-h period.

SK-OV-3 sphere cells (1×10^5) treated with or without $\gamma\delta$ T cells were suspended in aqueous buffer and stained with 0.0075 µg of anti-human HLA-DR PE (bioscience). The HLA expression was analyzed by FC500 flow cytometer.

Electron microscopy

SK-OV-3 adherent cells (differentiated cells), sphere cells treated with or without $\gamma\delta$ T cells, were collected for electron microscopy. In brief, samples were fixed in 2% glutaraldehyde for 2 h followed by osmication (1%) for 1 h and then allowed dehydration through a series of graded alcohols. This was followed by embedding in liquid Araldite before polymerization at 60°C overnight. They were then cut into 100-nm sections on copper electron microscope grids and stained with saturated aqueous uranyl acetate and 0.4% lead nitrate in 0.2 M sodium citrate, before being observed using a Philips CM80 electron microscope (Philips Electronics Nederland BV, Eindhoven, Netherlands).

ELISA for measurement of IL-17 production

The culture medium of SK-OV-3 sphere cells cocultured with or without $\gamma\delta$ T cells for 48 and 96 h was collected. After centrifugation, IL-17 in all the supernatants was measured by human IL-17 DuoSet ELISA Development System (R&D Systems, Minneapolis, MN, USA) according to the manufacturer's instructions.

In vivo xenograft experiments

All animal studies adhered to the protocols approved by the Institutional Animal Care and Use Committee of Shanghai Jiaotong University, Shanghai, China. The 500 dissociated spheroid SK-OV-3 cells combined with or without $\gamma\delta$ T cells (S:T ratio is 10:1) were counted, resuspended in 40 µl PBS and injected s.c. into the two sides of the flanks of 3- to 4-week-old female nude athymic mice (BALB/c-nu/nu;

Harlan). Engrafted mice were inspected biweekly for tumor appearance by visual observation and palpation until the tumor formed. Mice were killed by cervical dislocation. After mice were killed, the tumors' weight and size were measured, then fixed in 10% phosphate-buffered formalin and embedded in paraffin for sectioning (5 μ m) on a rotary microtome, followed by slide mounting, H&E staining and histologic assessment by a pathologist for tumor type and grade.

Statistics

Means, standard deviations, standard error and *P*-values (Student's *t* tests) were calculated using Microsoft Excel. *P*-values <0.05 were considered significant.

Results

The antitumor activity of $\gamma\delta$ T cells to ovarian cancer cell line SK-OV-3 cancer stem-like cells

Our previous data suggest that ovarian cancer is characterized by a pool of differentiated cells that comprise a majority of the total tumor cells and a small population of cells expressing the stem cell surface markers, such as Oct-4, Nanog, CD113, CD117, which is responsible for tumor initiation and maintenance [5, 6]. For the purpose of this study, we propagated ovarian cancer cell line SK-OV-3 spheres under serum-free conditions (Fig. 1a) and SK-OV-3 adherent cells under differentiating conditions (Fig. 1b). These cancer spheres were identified through the expression of CD133, CD117 and CD44. The flow cytometric profiles showed that the percentage of CD133⁺/CD117⁺ expressing cells in sphere cells (87.23%) is much higher than that in differentiated cells (19.78%, *P* < 0.01, Fig. 1c).

We then generated $\gamma\delta$ T cells from healthy donors. The ability of $\gamma\delta$ T cells to kill ovarian CSCs was assessed before and after treatment of $\gamma\delta$ T cells for 1 week. The $\gamma\delta$ T cells were cocultured with the sphere cells in a transwell system that was placed above the sphere cells layer (Fig. 2a). The sphere cells and $\gamma\delta$ T cells shared the same medium, but no direct cell–cell interactions are possible due to the physical separation of the cells by a polycarbonate membrane. The pore size of the transwell (0.4 μ m) allows no cell migration through the membrane. The $\gamma\delta$ T cells were able to exert significant cytotoxicity on the SK-OV-3 cancer sphere cells, the proliferation rate of the cancer sphere cells decreased to 40% after 48-h coculture with $\gamma\delta$ T cells, at a S:T ratio (SK-OV-3 sphere cells/ $\gamma\delta$ T cells) of 10:1 or 20:1 (Fig. 2b). Thus, the SK-OV-3 CSCs are highly susceptible to $\gamma\delta$ T-cell killing.

The $\gamma\delta$ T cells sensitize SK-OV-3 cancer stem-like cells to chemotherapy

It has been proposed that human $\gamma\delta$ T cells have several distinct pathways for antitumor immunity and the cytotoxicity to tumor cells can rely on secretion of proinflammatory cytokines and proapoptotic molecules or cell-contact-dependent lysis through a NK-like pathway or a TCR-dependent pathway [16]. With this no direct cell–cell interaction transwell system, after 48-h coculture with $\gamma\delta$ T cells (at a S:T ratio of 10:1), the real-time qPCR showed that the expression of stem cell marker genes (such as Nanog, Oct4 and sox-2) in sphere cells was decreased. The expression of ABCG2, Topo2a and Topo2b was also downregulated in treated sphere cells (Fig. 3a). This finding suggests that $\gamma\delta$ T cells could sensitize the ovarian CSCs to chemotherapy.

To examine whether the $\gamma\delta$ T cells increased the chemosensitivity of the SK-OV-3 sphere cells, we assessed the drug resistance properties expected for the sphere-forming cells cocultured with $\gamma\delta$ T cells by MTT assay. We found that the sphere cells possess strong chemo-resistant capacities. The inhibition rate of SK-OV-3 sphere cells with cisplatin and paclitaxel was 36 and 48%, respectively, which is consistent with our previous report. After 48 and 96 h cocultured with $\gamma\delta$ T cells, the inhibition rates of SK-OV-3 sphere cells with cisplatin and paclitaxel for 48 and 96 h were 78 and 89%, respectively (Fig. 3b, *P* < 0.01). In contrast, the inhibition rate of sphere cells cocultured with $\gamma\delta$ T cells alone was similar to that inhibition rate of sphere cells with cisplatin and paclitaxel (Fig. 3b, *P* > 0.05). Thus, the $\gamma\delta$ T cells increased the sensitivity of SK-OV-3 CSCs to chemotherapeutic drugs.

The $\gamma\delta$ T cells enhance HLA-DR antigen expression on SK-OV-3 cancer stem-like cells

The HLA molecules are membrane-bound transporters that carry peptides from the cytoplasm to the cell surface for surveillance by circulating T lymphocytes. Many types of tumor cells release low amounts of HLA molecules, allowing them to effectively evade the anticancer immune response of T cells [17, 18]. We analyzed the HLA-DR antigen, B7-1 and B7-2 in SK-OV-3 sphere cells by real-time qPCR and further confirmed that these sphere cells expressed low level of antigens. When tumor cells were cocultured with $\gamma\delta$ T cells in a transwell petri dish, the expression of HLA-DR, B7-1 and B7-2 antigen on tumor cells was increased significantly when compared with untreated sphere cells (*P* < 0.01, Fig. 3a). Further analysis by flow cytometry indicated that the expression of HLA-DR antigen on tumor cells was increased in a time-dependent manner (*P* < 0.01, Fig. 4).

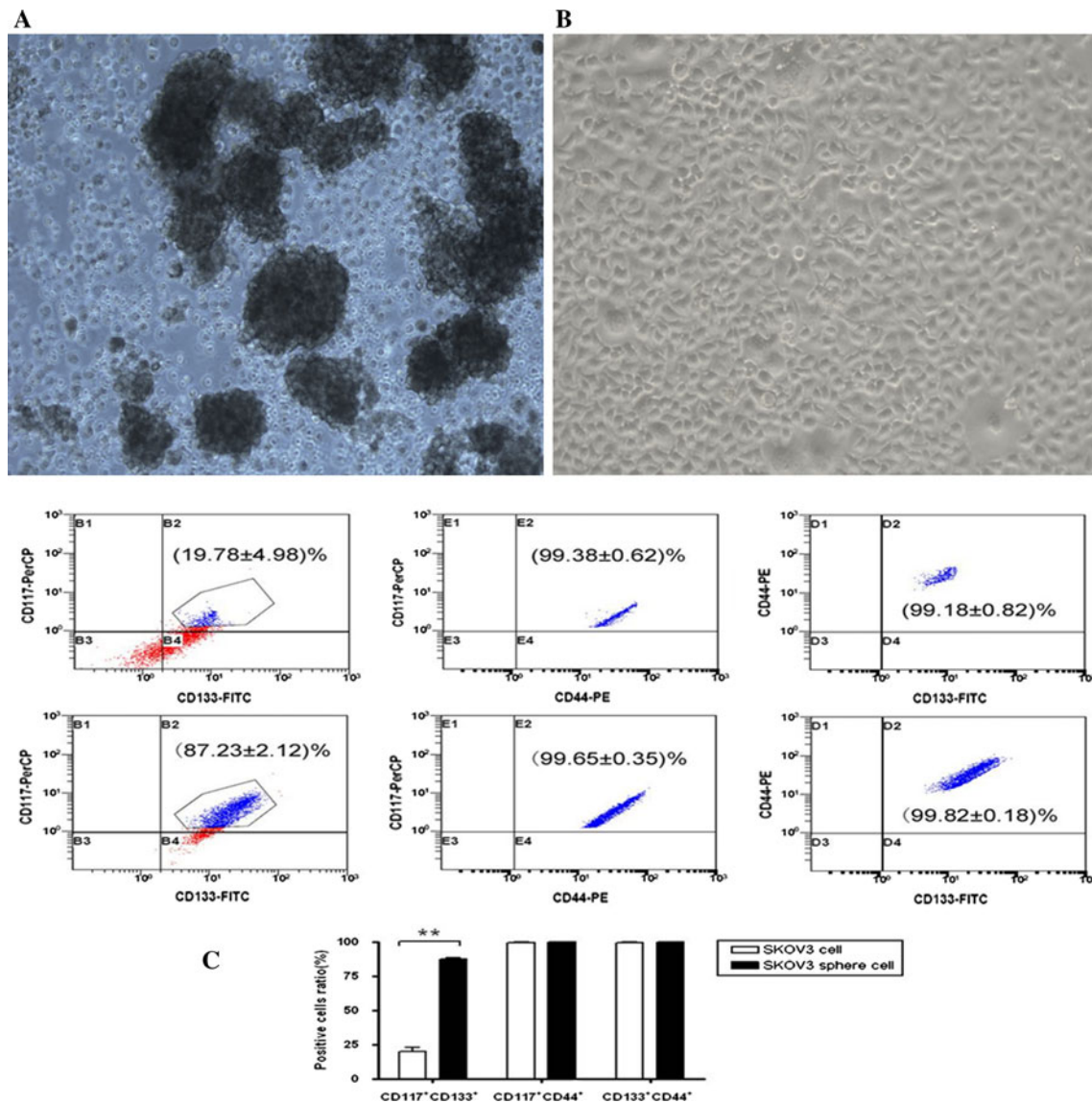


Fig. 1 SK-OV-3 CSCs express CD133 and CD117 markers. **a** Ovarian cancer cell line SK-OV-3 sphere cells. **b** SK-OV-3 differentiated cells. **c** Flow cytometric profiles of disaggregated SK-OV-3 sphere and differentiated cells. The percentage of CD133⁺/CD117⁺

expressing cells in sphere cells (87.23%) is much higher than that in differentiated cells (19.78%) (** $P < 0.01$), while CD117/CD44 or CD133/CD44 expression in sphere cells is the same as that in differentiated cells ($P > 0.05$)

$\gamma\delta$ T cells induced G2/M phase cell cycle arrest and subsequent apoptosis in SK-OV-3 cancer stem-like cells

The effects of $\gamma\delta$ T cells on the cell cycle progression of SK-OV-3 CSCs were investigated. The SK-OV-3 sphere cells were cocultured with $\gamma\delta$ T cells for 48–96 h and analyzed for cell cycle distribution by means of flow cytometry. Cultivation with $\gamma\delta$ T cells, SK-OV-3 sphere cells increased the population of cells in the G2 phase, with a reduction of cells in the G1 and S phase (Fig. 5). The SK-OV-3 CSCs apoptosis induced by $\gamma\delta$ T cells was further confirmed with flow cytometry. To quantify the percentage of SK-OV-3 CSCs in apoptosis caused by $\gamma\delta$ T cells, the

sphere cells with $\gamma\delta$ T-cell treatment for 48 and 96 h were subjected to annexin V/PI-double staining. Annexin V-positive cells were dramatically increased in a time-dependent manner, indicating that $\gamma\delta$ T-cell-induced apoptosis of SK-OV-3 sphere cells had occurred (Fig. 6). These results indicate that $\gamma\delta$ T cells led SK-OV-3 sphere cells to cell cycle arrest at the G2 phase followed by apoptosis.

The apoptotic morphology of SK-OV-3 sphere cells cocultured with $\gamma\delta$ T cells

We first investigated the ultrastructure of SK-OV-3 sphere cells. Under electron microscopy, the nuclear/cytoplasmic ratio of the sphere cells was lower than that of

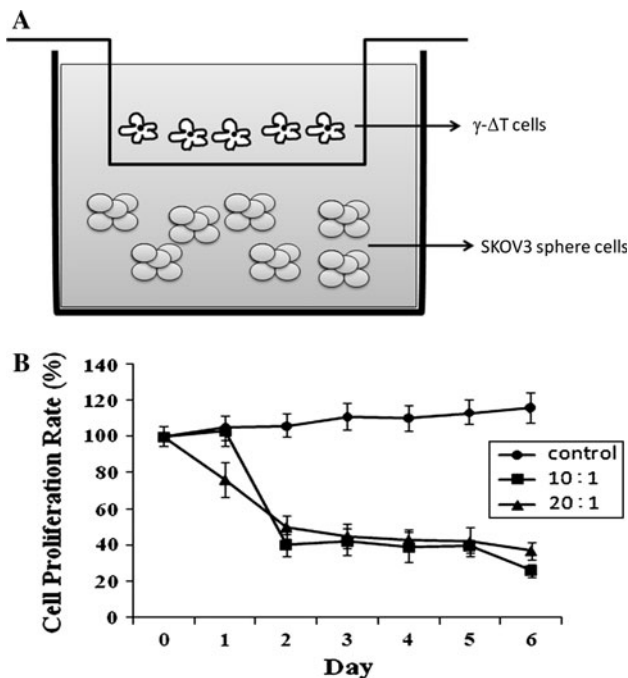


Fig. 2 SK-OV-3 cancer stem cells cocultured with $\gamma\delta$ T lymphocytes. **a** Schematic representation of the transwell coculture system. SK-OV-3 sphere cells cocultured with $\gamma\delta$ T lymphocytes by no cell contact, cell to cell communication with soluble proteins. **b** The proliferation rates of SK-OV-3 cancer stem cells cocultured with $\gamma\delta$ T lymphocytes are lower than those of CSCs treated with medium alone ($P < 0.001$). There is no difference in the cytotoxic activity of $\gamma\delta$ T lymphocytes against SK-OV-3 cancer stem cells between an S:T ratio of 10:1 and 20:1 ($P > 0.05$). S:T ratio is the number of SK-OV-3 sphere cells versus the number of $\gamma\delta$ T cells

the differentiated cells. Compared with the differentiated cells, the sphere cell's nucleus becomes smaller, and the sphere cells had much more cytoplasmic organelles that formed by mitochondria and Golgi (Fig. 7a–c). The numerous secretory granules in sphere cell suggest that ovarian cancer stem cell has the ability to secrete large amounts of cytokine.

Flow cytometry results showed that $\gamma\delta$ T cells induced cell apoptosis in sphere cells. Scanning electron microscopic analysis provided further morphologic insights into the apoptotic process. The apoptotic bodies were seen “shelled out” from the bounding membrane, accompanied by chromatin condensation and margination in nucleus (Fig. 7d). These observations suggest that cell apoptosis operates when the sphere cells were exposed to the coculture system with $\gamma\delta$ T cells.

IL17 production significantly increases after cocultured with $\gamma\delta$ T cells

Recent studies demonstrated that IL-17 production by $\gamma\delta$ T cells played a key role in the antitumor effect [19]. In this

study, we examined IL-17 levels in the culture medium of SK-OV-3 sphere cells after cocultured with $\gamma\delta$ T cells. As shown in Fig. 8, compared with control, IL-17 production was significantly increased after the SK-OV-3 sphere cells cocultured with $\gamma\delta$ T cells for 48 and 96 h (Fig. 8, $P < 0.01$). Therefore, IL-17 levels in the coculture system might be produced by $\gamma\delta$ T cells.

Reduction of tumor burden by $\gamma\delta$ T cells in xenograft mouse model

We developed an ovarian cancer xenograft mouse model to explore the in vivo efficacy of $\gamma\delta$ T cells. Five hundred disaggregated tumor sphere cells were combined with $\gamma\delta$ T cells (S:T ratio of 10:1), and then, they were injected s.c. into nude mice. Tumors were monitored twice a week for 2 months. Control animals received tumor cells only. All the tumors were formed with an average of 58 days latency, and only one of six mice treated with $\gamma\delta$ T cells formed tumor (Fig. 9a, b). The average weight of tumor in the control was much higher than that in $\gamma\delta$ T-cell-treated group ($3,255 \pm 34$ mg vs. 96 ± 4 mg, $P < 0.0001$, Fig. 9d). All subcutaneous xenograft tumors were categorized as serous adenocarcinoma of Grade 2/3, which was the original tumor phenotype of the SK-OV-3 cell line (Fig. 9c). The $\gamma\delta$ T cells significantly reduced the tumor burden in mouse xenograft model.

Discussion

Ovarian CSCs have been identified in established ovarian cancer cell line and patient-derived ovarian tumor samples [3–6]. Ovarian CSC-bearing stem cell markers can be identified and found to possess high tumorigenicity. Furthermore, these cancer-initiating cells are more resistant to carboplatin, cisplatin and paclitaxol, three commonly used drugs in the treatment of ovarian cancer. The chemo-resistance in CSCs may go through various mechanisms. One of them suggested that the ATP-binding cassette ABC transporter proteins, which are highly expressed in CSCs, contribute to drug resistance by pumping the drugs out of the cells and reducing intracellular levels and diminishing the ability of these agents to cause tumor cell death [20, 21]. Recently, another mechanism of treatment resistance has been proposed, which suggests that the low level of reactive oxygen species (ROS) in CSCs is related to radioresistance [22]. Thus, CSCs may be important factors in the development of treatment resistance and in the relapse of cancers. Previously, we reported that the nonadherent spheres cells isolated from the primary ovarian tumor samples or SK-OV-3 cell line display distinct phenotypic/genotypic characterization from differentiated cells by cDNA

Fig. 3 $\gamma\delta$ T cells increase the chemo-sensitivity of SK-OV-3 sphere cells. **a** As shown by real-time qPCR, $\gamma\delta$ T cells inhibit the expression of cancer stem cell genes and increase the antigen expression of SK-OV-3 sphere cells ($P < 0.01$, 18S RNA was used as an internal control). **b** After 48 and 96 h cocultured with $\gamma\delta$ T cells at a S:T ratio of 10:1, the inhibition rates of SKOV3 sphere cells with cisplatin (CDDP, 40 μ Mol) and paclitaxel (PTX, 10 μ Mol) were increased by MTT assay ($P < 0.01$)

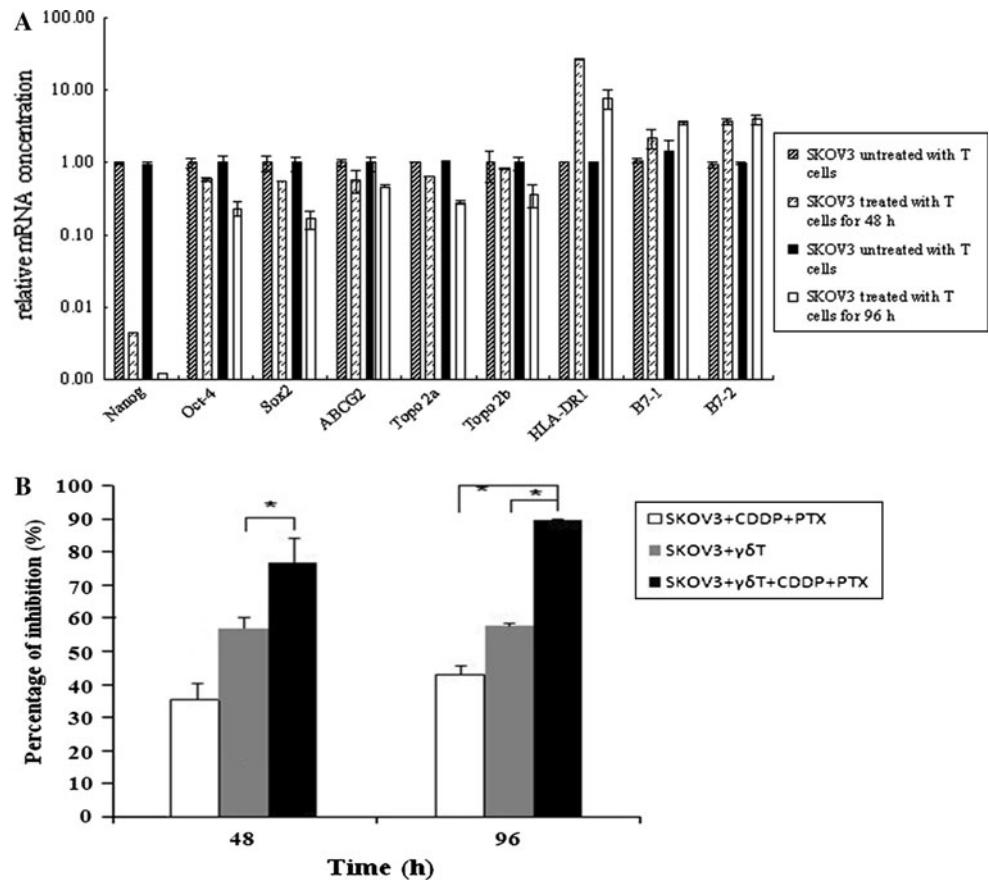
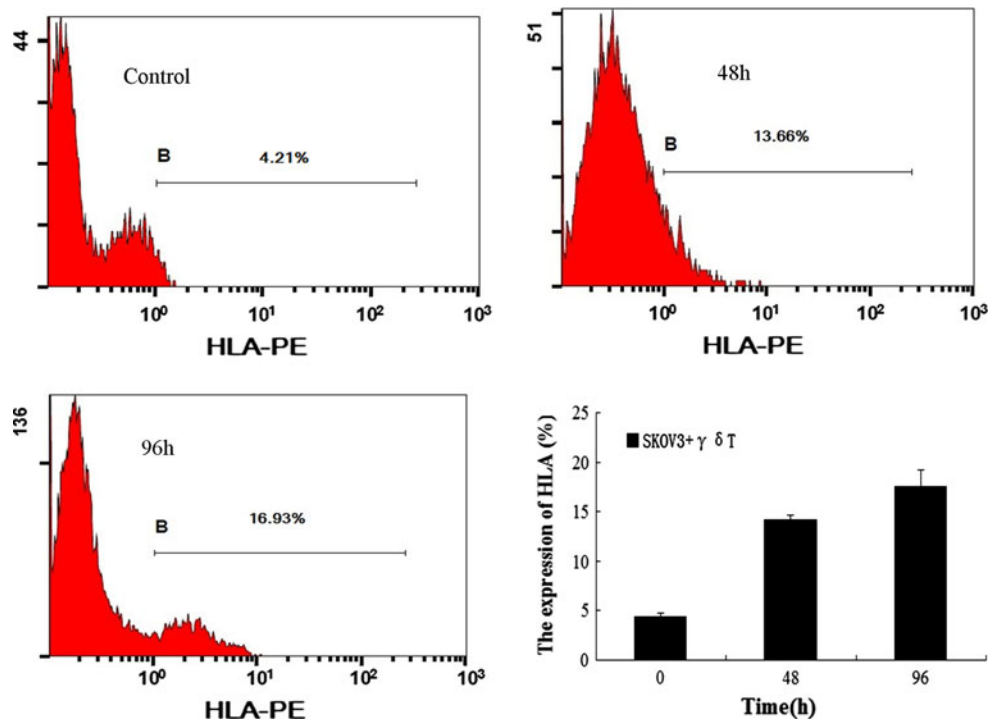


Fig. 4 The expression of HLA-DR antigen on SK-OV-3 sphere cells is increased after being treated with $\gamma\delta$ T cells for 48 and 96 h as indicated by flow cytometry ($P < 0.01$)



microarray [5, 6]. Cell lines are a useful tool to analyze molecular cell markers and cellular behavior under controlled experimental conditions. Isolation of this subpopulation from

the SK-OV-3 cell line under stem cell culture conditions affords a suitable in vitro model for identifying and understanding the criteria of CSCs.

Fig. 5 Cell cycle analysis of SK-OV-3 CSCs cocultured with $\gamma\delta$ T cells. SK-OV-3 sphere cells were cocultured with $\gamma\delta$ T cells for 48 and 96 h and then stained with PI. The DNA content was analyzed by means of flow cytometry. After 48- and 96-h treatment with $\gamma\delta$ T cells, the proportion of SK-OV-3 sphere cells in the G2/M phase was increased compared with that in the untreated SK-OV-3 sphere cells ($P < 0.01$). Data shown are triplicate experiments

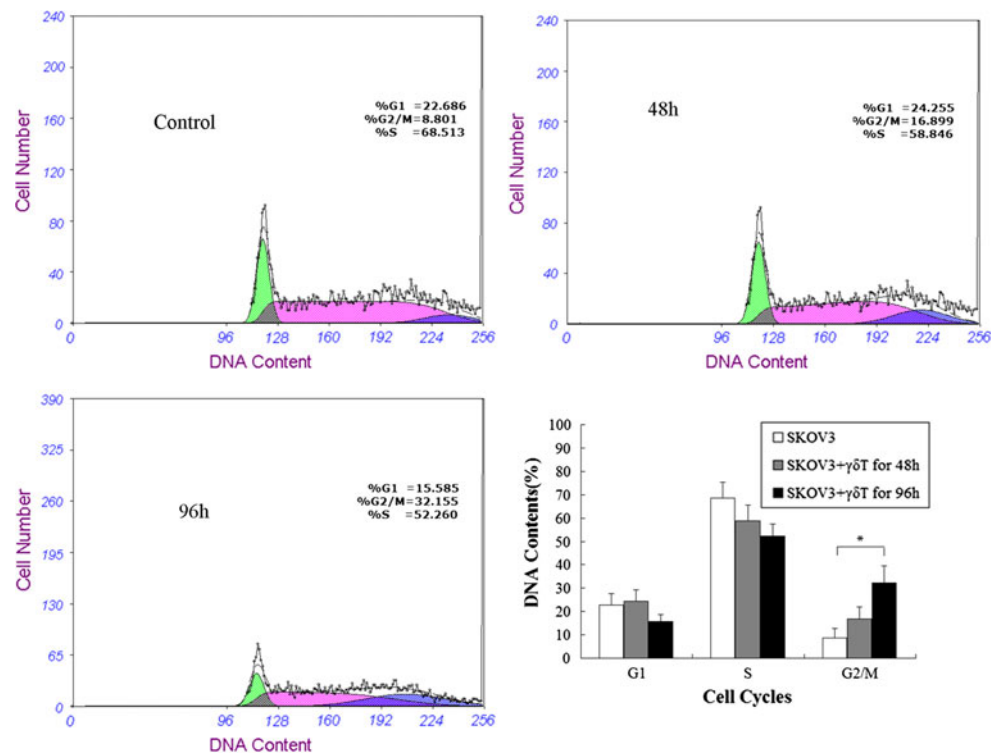
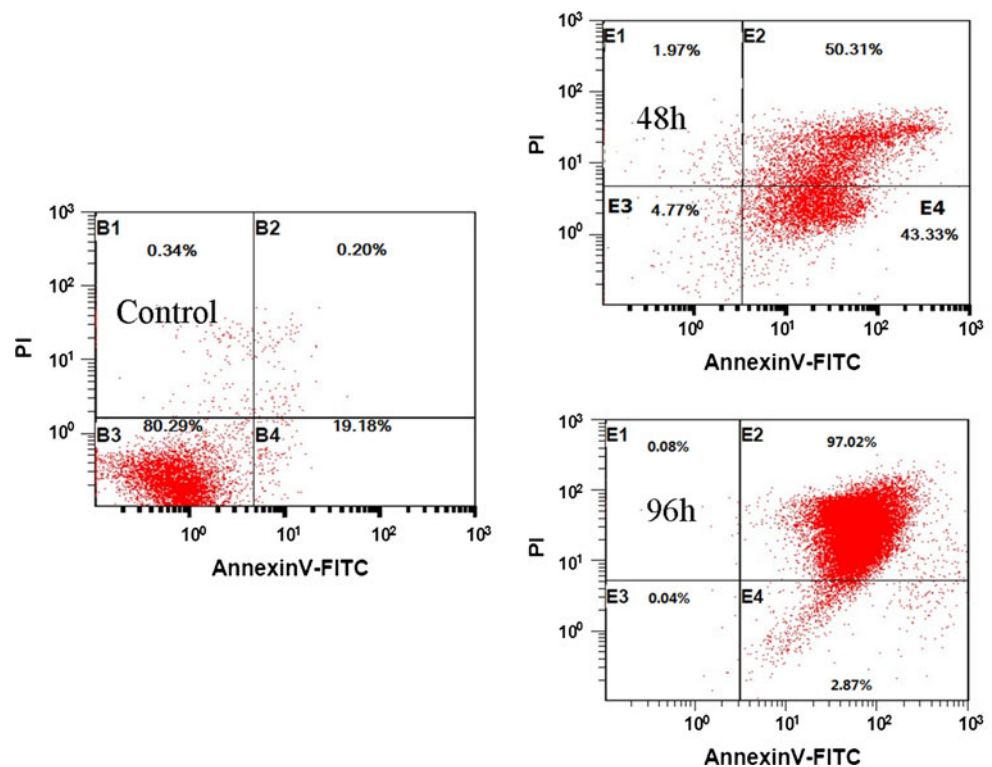


Fig. 6 Detection of apoptotic cells by annexin V and PI-double staining. SK-OV-3 sphere cells were cocultured with $\gamma\delta$ T cells for 48 and 96 h, stained with annexin V-FITC and PI labeling and analyzed by flow cytometry. The percent digits refer to the annexin V-positive cells. Three independent experiments were performed, and all gave similar results



Therefore, it would be desirable to develop therapies that specifically kill CSCs. The question is how immunosystems recognize and kill ovarian CSCs?

So far, two subsets of T cells have been defined: $\alpha\beta$ T cells and $\gamma\delta$ T cells. Human $\gamma\delta$ T cells constitute only 2–5%

of circulating lymphocytes. Peptide-major histocompatibility complex-specific T-cell responses are known to be achieved by $\alpha\beta$ T cells, while $\gamma\delta$ T cells exhibit potent MHC-unrestricted lytic activity against different tumor cells, suggesting an important role of $\gamma\delta$ T cells in defense.

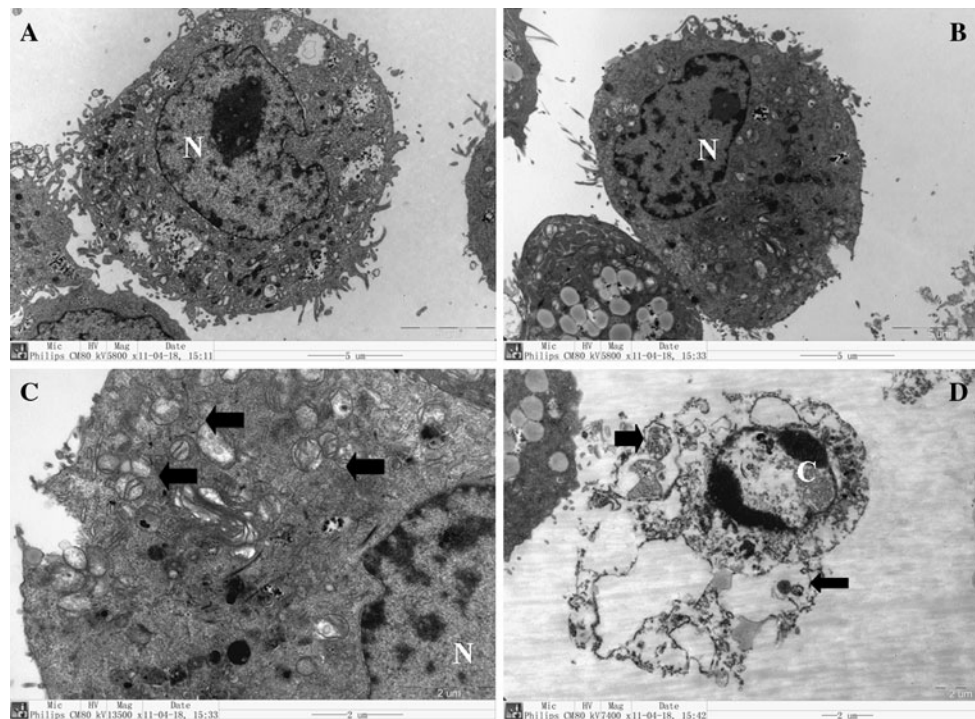


Fig. 7 Ultrastructure of SK-OV-3 cells. **a** Electron microscopic morphology of differentiated cell. *N* nucleus. **b** Electron microscopic morphology of sphere cell. *N* nucleus. **c** The sphere cell has much more

cytoplasmic organelles (*arrows*). **d** The apoptotic sphere cell induced by $\gamma\delta$ T cell. The apoptotic bodies contain well-preserved organelles (*arrows*) with chromatin margination (*c*)

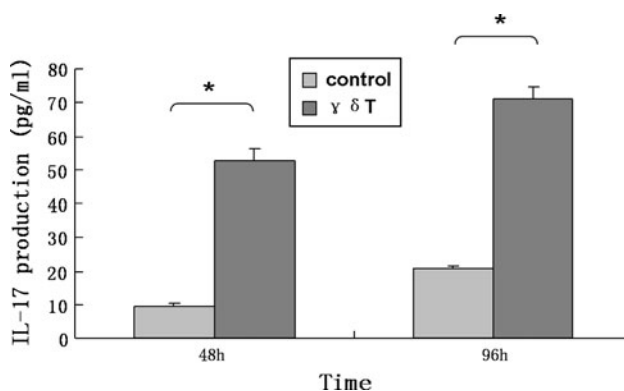


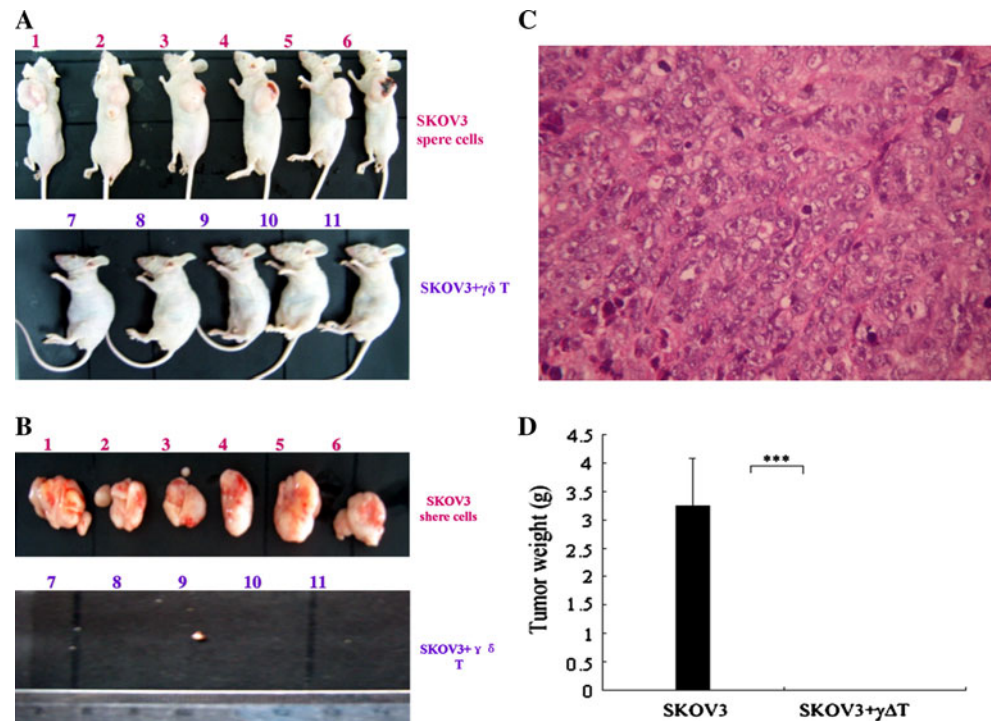
Fig. 8 The levels of IL17 production were assessed in the supernatants of SK-OV-3 sphere cells cocultured with $\gamma\delta$ T cells for 48 and 96 h (Representative data of three separate experiments are shown, $P < 0.01$)

Accumulating evidence shows that $\gamma\delta$ T cells have antitumor effects both in vitro and in animal models. Recently, researchers have tested their function in clinical therapies in patients. The count of $\gamma\delta$ T cells is correlated with the patients' clinical survival rate. Tanaka and colleagues reported that in advanced renal cell carcinoma, a higher survival rate after surgery is correlated to higher $\gamma\delta$ T-cell frequency in patients' peripheral blood [23]. Several other studies showed that $\gamma\delta$ T cells are able to recognize and kill various differentiated tumors cells, such as glioblastoma, myeloma, colon cancer and lymphoma cell lines [24–27].

Our strategy is based on the observation that tumorigenicity of cancer cells is not created equal and only a subpopulation of tumor cells are endowed with capacity to initiate tumor and implicated in chemo-resistance. We hypothesize that $\gamma\delta$ T cells targeting this subpopulation of ovarian cancer cells may have significant therapeutic value. Ovarian cancer stem cells that have tumor-specific antigens would be an attractive target for immunotherapy. Weng et al. [28] isolated a subset of ovarian cancer cells with a CD44⁺ phenotype in samples from ovarian cancer patients, using fusions of dendritic cells and CSCs that could specifically target the CD44⁺ subpopulations. Thus, the tumor-specific antigens expressed by CSCs can be recognized by cytotoxic T lymphocytes (CTLs). In this study, we isolated the SK-OV-3 sphere cells that expressed stem cell markers and were resistant to chemo-agents. These cells, which were referring to cancer stem-like cells, were cocultured with $\gamma\delta$ T cell. We show that $\gamma\delta$ T cells are able to efficiently induce the apoptosis of ovarian sphere cells and increase the sensitivity of SK-OV-3 sphere cells to chemotherapeutic drugs. Dramatically, the $\gamma\delta$ T cells significantly reduced the tumor burden in mouse xenograft model.

Todaro et al. [29] recently isolated colon CSCs from primary colon cancer tissues under serum-free culture conditions and evaluated their susceptibility to $\gamma\delta$ T cells. The CSCs/TICs from colon cancer tissues showed susceptibility to $\gamma\delta$ T cells. A small nonpeptidic phosphorylated compound

Fig. 9 Robust in vivo propagation of human ovarian tumors in nude mice by SK-OV-3 sphere cells combined with $\gamma\delta$ T cells or sphere cells alone. **a** Xenograft tumor formed after injection of 500 sphere cells, only one tumor formed in the $\gamma\delta$ T-cell treatment group. **b** The different tumor size in two groups. **c** Representative H&E staining sections of tumors were classified as advanced Grade 2/3 serous adenocarcinoma, which was the original tumor phenotype of the SK-OV-3 cell line. **d** The average tumor weight in the control was much higher than that in $\gamma\delta$ T-cell group ($P < 0.0001$)



(Zoledronate) sensitized colon cancer CSCs/TICs to $\gamma\delta$ T cells. These data suggest that $\gamma\delta$ T-cell cytotoxicity to CSC was mainly TCR mediated, whereas NKG2D played a role only when tumor targets expressed several NKG2D ligands. However, we revealed that the recognition and efficient killing of ovarian cancer stem-like cells by $\gamma\delta$ T cells may rely on cell-contact-independent pathway.

$\gamma\delta$ T cells produce a series of cytokines in pathology, including interferon (IFN)- γ , tumor-necrosis factor (TNF)- α and IL-17, which are critical for pathogen elimination, immune regulation and autoimmunity [30–32]. $\gamma\delta$ T cells are the major source of IL-17 production in the antitumor effects [19]. In order to elucidate the interactions of ovarian CSC and $\gamma\delta$ T cells, we have established a transwell system that permits cell-contact-independent communication through diffusible soluble factors only. Although human $\gamma\delta$ T cells have several distinct pathways for antitumor immunity, our results showed that cytokines secreted, such as IL-17, by $\gamma\delta$ T cells play a major role in the immunotherapy targeting ovarian cancer stem cells. Although defining the ligands of $\gamma\delta$ T cells is still far from the reach, cytokines production by $\gamma\delta$ T cells may open the window for understanding the whole picture of immune response to CSC.

Moreover, tumor cells express less HLA class II and display poor costimulatory molecule expression (such as B7-1 and B7-2), which leads them to escape immune surveillance. For this reason, genetically modified tumor cells expressing costimulatory molecules have been used as cancer vaccines in both experimental tumor models and in clinical trials [33]. In this study, we found that the $\gamma\delta$ T

cells induced the expression of HLA-DR, B7-1 and B7-2 in sphere cells, and the expression of HLA-DR antigen on the sphere cells was increased in a time-dependent manner by flow cytometry assay. These findings suggest that $\gamma\delta$ T cells may promote the antigen expression of tumor cells, and it is helpful for the immune system to identify tumor cells.

In conclusion, the in vitro expansion of $\gamma\delta$ T cells may play an important role in tumor immunotherapy, and these cells combined with other antitumor agents may be of significant clinical benefit in the treatment of ovarian cancer. CSC-targeted immunotherapy could represent a noncross-resistant strategy that complements conventional therapy of ovarian cancer.

Acknowledgments This work was supported by grants from Shanghai Municipal Council for Science and Technology (No. 09411968300) and the Key Project Fund of Shanghai Municipal Health Bureau (2010011).

Conflict of interest The authors declare that they have no financial conflict of interest.

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